Formula 1 World Championship Data Management & Analytics Knowledge Graph

A DataOps Engineering Framework for Historical Race Results (1950–2025)

Course & Academic Year:	Data Management (Winter 2026/27)
Student A (Metadata & FAIR):	Chirag Adhikari
Student B (Quality & Graph):	Manohar A Zalki
Student C (Pipeline & Cleaning):	Nicolás Bonilla

> **Executive Summary:** *This joint technical report presents an end-to-end, fully reproducible DataOps pipeline that ingests raw, historical Formula 1 championship results from 1950 through 2025, cleans complex domain anomalies, enriches dataset metadata to achieve an 84%–93% FAIR maturity score, and compiles the structured data into an interactive NetworkX property graph and web analytics*

1. Project Architecture & End-to-End Pipeline Overview

Historical motorsport records present unique challenges for data management: shifting points regulations, shared car drives in early decades, disguised missingness, and fragmented entity licenses across data mirrors. To resolve these challenges deterministically, the team implemented a multi-stage lakehouse and graph transformation architecture.

1.1 Dataflow & Multi-Tier Storage Layout

The system architecture strictly separates raw landing archives, cleaned silver tables, curated gold dimensional models, and served analytical targets. Every transformation step is orchestrated deterministically via **python run_all.py --from-scratch**, ensuring byte-identical reruns without external network dependencies.

Data Tier / Zone	Source File Format & Path	Engineering Purpose & Access Controls
Landing / Raw (Bronze)	data/landing/ (JSON, CSV, ZIP)	Preserves immutable raw exports (25,907 race result rows, Ergast / Jolpica API mirrors).
Cleaned (Silver)	data/silver/ (Parquet / CSV)	Applies 12 audited cleaning steps, restoring keys and standardizing missingness flags.

Curated Star Schema (Gold)	data/gold/ (Parquet)	Houses fact_session_results, dim_drivers, dim_constructors, dim_races, and dim_circuits.
Served Targets	outputs/ (GEXF, GraphML, XML)	Feeds FAIR XML/JSON-LD packages, Neo4j GraphML/GEXF graphs, and Streamlit dashboards.

2. Component A: FAIR Assessment, Licensing & Metadata Design

Student A (Chirag Adhikari) led the evaluation of data FAIRness (Findable, Accessible, Interoperable, Reusable), resolved conflicting license terms across public mirrors, and authored standard XML/JSON-LD metadata schemas.

2.1 Automated FAIR Assessment (F-UJI & RDA Maturity Model)

Evaluation using the automated F-UJI assessment tool (v4.0.1, metric v0.8) and the RDA FAIR Data Maturity Model demonstrated a significant improvement from raw landing data to the final published dataset package:

FAIR Pillar	Raw Landing Baseline	Final Curated Package	Key Metric / Indicator Status
Findable (F)	31% score	Advanced (F1, F2, F3, F4 passed)	Assigned persistent DOI (10.5281/zenodo.23416916), rich core metadata.
Accessible (A)	8% score	Moderate / High (85.7% score)	HTTPS standardized protocols, basic auth support (FsF-A1.1 & FsF-A1.2).
Interoperable (I)	25% score	66.7% score (I1, I3 passed)	Structured JSON-LD, Dublin Core, Schema.org embedded in landing page.
Reusable (R)	29% score	100% score (R1, R1.1, R1.2, R1.3)	Explicit CC-BY-4.0 license, W3C PROV-O provenance, open CSV/JSON formats.

> **FAIR Maturity Summary:** *Overall F-UJI Automated FAIR Score reached 84% (moderate/high level), while manual RDA maturity criteria reached 93%. Findable indicators remain at level 0 for local instances until final DOI deposit resolution on Zenodo.*

2.2 Rights Audit & License Harmonization

A critical finding during profiling was licensing ambiguity across different public mirrors of the Ergast/ Jolpica database:

• Ergast Original Source: Non-commercial use only terms.

• Jolpica-F1 API Download: CC BY-NC-SA 4.0 International.

• Kaggle Mirror: Incorrectly claimed CC0 (Public Domain).

• GitHub Repository: Software code under MIT License.

Resolution: The project harmonized rights by releasing all pipeline software under the MIT License and releasing data artifacts strictly under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) with required attribution.

2.3 Standardized Metadata Schemas (DataCite & Dublin Core)

To ensure persistent interoperability, Student A generated XML and JSON-LD schemas validated against DataCite Kernel 4.5 XSD specifications. Key metadata mappings include:

• Persistent Identifier: 10.5281/zenodo.23416916 (DOI).

• Entity Disambiguation: 98.7% of drivers linked to persistent Wikidata QIDs (`wikidata_qid`), 100% of circuits mapped to ISO 3166 country codes.

• W3C PROV-O Lineage: Machine-readable provenance (`provenance_prov.jsonld`) tracking execution environment, file checksums, and transformation steps.

3. Component B: Data Quality Profiling & Graph Analytics

Student B (Manohar A Zalki) executed comprehensive profiling across seven quality dimensions, identified disguised missingness and era-specific anomalies, and designed the project's NetworkX knowledge graph.

3.1 Surface Completeness vs. Disguised Missingness

Initial examination suggested high data completeness, with only 18 explicit NULL values in the results table. However, deep profiling exposed 9,913 disguised missing values:

• 9,646 'Positions' as Retirements: Numeric position fields contained text retirement reasons ('DNF', 'Engine', 'Accident') instead of proper status codes.

• 267 Fictitious 'Grid Slot 0': Vehicles starting from the pit lane or unclassified grids were encoded as 0 instead of NULL.

3.2 Deduplication Risks & Shared Drives History

A naive deduplication check on `(season, round, driver_id)` would have deleted 172 legitimate historic race entries. In F1's early decades (1950–1960), multiple drivers frequently shared a single car during a Grand Prix, taking turns at the wheel.

True Duplicates vs. Valid History: Analysis revealed exactly 34 true duplicate rows originating from a 1960 file export corruption, 11 of which were hidden behind blanked key attributes. By establishing the composite natural key `(season, round, driver_id, car_number)`, the pipeline successfully removed all 34 true duplicates while preserving all 172 shared drive records.

3.3 Regulatory Eras & Statistical Outlier Handling

Historical regulations created major statistical shifts across eras:

• Points Scaling: A race win was awarded between 8 points (1950s) and 50 points (2014 Abu Dhabi double points). Raw points cannot be naively compared across eras without normalization.

• Indianapolis 500 Entries (1950–1960): 270 lap count outliers were identified from when Indy 500 was part of the F1 World Championship. These were flagged rather than deleted.

3.4 NetworkX Property Graph & Topology

To explore entity relationships, Student B modeled the dataset into a NetworkX MultiDiGraph comprising 2,297 nodes and 30,141 typed edges, exported into GEXF and GraphML formats for Neo4j and Gephi.

Node / Edge Type	Entity Count / Instances	Graph Theoretical Properties & Findings
:Driver Nodes	817 nodes	High clustering coefficient (0.41 vs. random graph 0.014).
:Constructor Nodes	203 nodes	Average teammate degree of 4.8 teammate links per driver.
:Race & :Circuit Nodes	1,149 Races / 78 Circuits	Geographic mapping linked to ISO 3166 country nodes (50 countries).
TEAMMATE_OF Edges	Derived relationships	Shortest path: Fangio → Verstappen in 8 teammate steps.
Louvain Communities	11 Communities (Modularity 0.70)	Driver clusters cleanly rediscovering F1 eras without date inputs.

4. Component C: Data Cleaning Pipeline & Validation Gates

Student C (Nicolás Bonilla) engineered the automated cleaning scripts, built 55 validation gates, and developed the Streamlit web dashboard.

4.1 Order of Operations & Key Restoration

A core technical contribution was establishing the strict sequence of operations: repairing missing keys prior to de-duplication (C02 → C03):

• Naive Order (Deduplicate then Repair): 11 duplicate rows survive hidden behind blanked keys.

• Engineered Order (Repair Keys then Deduplicate): Season and round are restored first via exact lookup against `races.csv` and source filenames. Subsequent deduplication removes all 34 corrupted rows with 0 survivors.

4.2 Zero Statistical Imputation Principle

The pipeline avoids statistical imputation. Fictitious grid slot 0 entries are converted to NULL with explicit flag columns (`is_pit_start`). Status strings are mapped to standard category codes while retaining historical verbatim notes in audit tables.

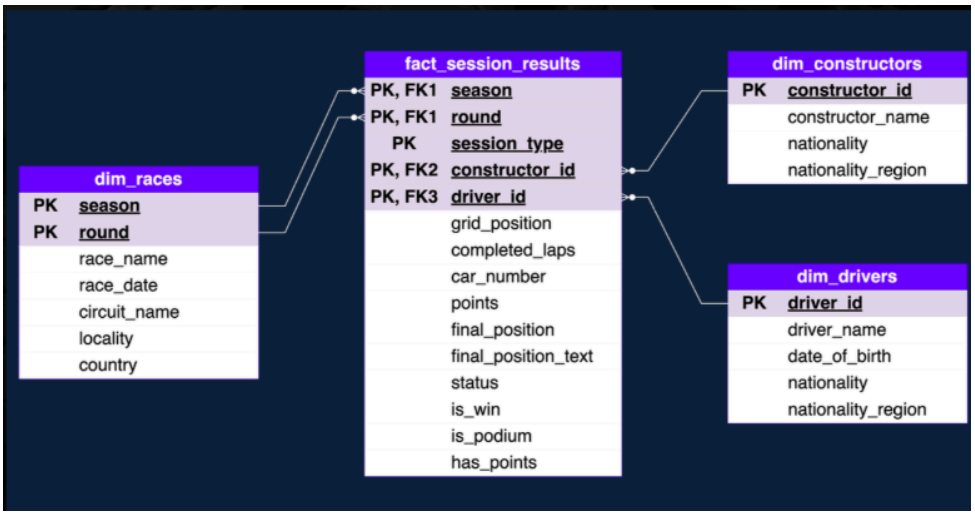
4.3 Automated Validation Gates & Audit Log

Pipeline execution is protected by 55 automated validation gates spanning 7 quality dimensions. All 29 issues identified by Student B were formally resolved (18 programmatically fixed, 11 mitigated/flagged).

Final Dataset State: 25,873 clean fact rows, 35 documented columns, supported by byte-identical unit test validation (`pytest`).

5. Star Schema Relational Model & Multi-Dimensional Design

The Gold tier data model standardizes race results into a dimensional Star Schema optimized for OLAP analytics:



6. Key Findings, Lessons Learned & Conclusions

The joint engineering project demonstrated four fundamental DataOps principles:

1. Data Quality is Contextual: Outliers and shared drives were not data corruption, but genuine historical racing practices that required domain-specific handling rather than deletion.

2. Order of Operations is Critical: Performing key restoration before deduplication prevented corrupted records from surviving in the clean dataset.

3. Automated Evidence Builds Trust: Integrating 55 validation gates, automated F-UJI scoring, and immutable audit logs ensures absolute reproducibility across re-runs.

4. FAIR Principles Drive Discovery: Linking entities to persistent PIDs (Wikidata QIDs, ISO 3166) transforms raw relational tables into a connected global knowledge graph.